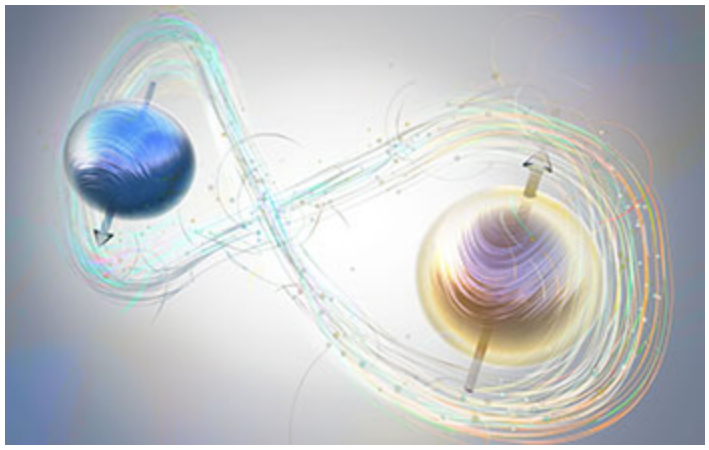
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Announcement

NSF establishes 3 new institutes to address critical challenges in quantum information science



Artist's rendition of quantum entanglement.
Credit: Nicolle R. Fuller/NSF

July 21, 2020

Quantum phenomena have puzzled and delighted scientists for over a century, revealing unique, counter-intuitive characteristics of matter like superposition and entanglement. For four decades, the U.S. National Science Foundation has worked to enable breakthroughs in quantum information science and engineering that harness what researchers have learned about quantum phenomena to develop technologies like quantum computers, sensors, and communications. These quantum technologies will have enormous consequences for the national and global economy. To unleash that potential, researchers must overcome several major, fundamental challenges in quantum information science and engineering.

With these unresolved questions in mind, NSF launched the Quantum Leap Challenges Institutes program. And today, NSF, in partnership with the White House Office of Science and Technology Policy, is announcing $75 million for three new institutes designed to have a tangible impact in solving these problems over the next five years.

These institutes are a central piece of NSF's response to key federal initiatives to advance quantum information science, including the National Quantum Initiative Act of 2018 and the White House's ongoing focus on American leadership in emerging technologies. Quantum Leap Challenge Institutes also form the centerpiece of NSF's Quantum Leap, an ongoing, agency-wide effort to enable quantum systems research and development.

"Quantum information science has the potential to change the world. But to realize that potential, we must first answer some fundamental research questions," said NSF Director Sethuraman Panchanathan. "Through the Quantum Leap Challenge Institutes, NSF is making targeted investments. Within five years, we are confident these institutes can make tangible advances to help carry us into a true quantum revolution."

"America's future depends on our continued leadership in the most cutting-edge industries of tomorrow. With the announcement of three new quantum institutes, the Trump Administration is making a bold statement that the United States will remain the global home for QIS research. Our new Quantum Leap Challenge Institutes will advance America's long history of breakthrough discoveries and generate critical advancements for years to come," said Michael Kratsios, U.S. Chief Technology Officer.

NSF is establishing three institutes:

- NSF Quantum Leap Challenge Institute for Enhanced Sensing and Distribution Using Correlated Quantum States <https://web.archive.org/web/20240520090951/https://www.nsf.gov/awardsearch/showAward?AWD_ID=2016244> . Quantum sensors that can measure everything from radiation levels to the effects of gravity will be more sensitive and accurate than classical sensors. This institute, led by the University of Colorado, will design, build, and employ quantum sensing technology for a wide variety of applications in precision measurement.
- NSF Quantum Leap Challenge Institute for Hybrid Quantum Architectures and Networks <https://web.archive.org/web/20240520090951/https://www.nsf.gov/awardsearch/showAward?AWD_ID=2016136> . Developing more robust quantum processors is a significant challenge in quantum information science and engineering. This institute, led by the University of Illinois, Urbana-Champaign, will build interconnected networks of small-scale quantum processors and test their functionality for practical applications.
- NSF Quantum Leap Challenge Institute for Present and Future Quantum Computing <https://web.archive.org/web/20240520090951/https://www.nsf.gov/awardsearch/showAward?AWD_ID=2016245> . Today's quantum computing prototypes are rudimentary, error-prone, and small-scale. This institute, led by the University of California, Berkeley, plans to learn from these to design advanced, large-scale quantum computers, develop efficient algorithms for current and future quantum computing platforms, and ultimately demonstrate that quantum computers outperform even the best conceivable classical computers.

The institutes comprise an interconnected community of 16 core academic institutions, 8 national laboratories, and 22 industry partners. Through integrating the perspectives and resources of multiple disciplines and sectors, they promote a sustainable ecosystem for innovation. In addition to their research, these centers will also make strides in training and educating a diverse, quantum-ready U.S. workforce. They will develop new in-person and online curricula for students and teachers at all educational levels, from primary school to professionals.

More information on NSF-supported quantum information science and engineering research can be found here <https://web.archive.org/web/20240520090951/https://www.nsf.gov/news/factsheets/Factsheet_Quantum-

proof7_508.pdf> .

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